

Why Water Should Boil at Minus 80

Fitting a trend line through three hydrides, extrapolating it back to the fourth, and reading what the miss is telling you.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/why-water-boils/

1 Before you open anything

Answer from what you already know. The point is to have a number on paper that the demonstration will later confirm or contradict.

PREDICTION

Three of the group 16 hydrides boil at these normal (1 atm) boiling points:

H₂S, 34.08 g/mol, -60.3 °C · H₂Se, 80.99 g/mol, -41.3 °C · H₂Te, 129.62 g/mol, -2.2 °C

Using only those three, predict the boiling point of H₂O (18.02 g/mol). Give a number in °C.

Write one sentence saying how you got it.

Water actually boils at 100.0 °C. Subtract your prediction from that. How large is the disagreement, in °C?

2 Setting up the control

Group 14 is the control series: nothing in it can hydrogen bond, so only London dispersion is acting. Establish what a well-behaved series looks like before you go looking for a broken one.

1. Open the demonstration and press **Control only**. Only the group 14 box should be ticked, and **Show extrapolation lines** should stay ticked.
2. Under **Inspect a group**, the tab **14** should be selected. Read the four boxes: *Lightest*, *Actual b.p.*, *Trend says*, *Gap*.

- Below the dividing line, read *Trend through the 3 heavy members* and *Fit quality* (R^2 , heavy members).
- Record all six values in the table.

Lightest	Actual b.p. (°C)	Trend says (°C)	Gap (°C)	Slope (°C per g/mol)	R^2

a. The dashed line was fitted to SiH_4 , GeH_4 and SnH_4 only — CH_4 was left out of the fit. Why does leaving it out matter for the argument being made?

b. The slope is positive. Name the only intermolecular force acting in this series, and say in one sentence why it strengthens down the group.

c. Hover or tap the SnH_4 point to open its tooltip. Record the molar mass and boiling point it reports, in both °C and K.

3 One control, four groups

The **Inspect a group** tabs are the control you are going to vary. Selecting a tab also switches that series on, so you can step straight through all four.

- Press **All four** so every series is on screen.
- Click tab **14**, then **15**, then **16**, then **17**. Each time, record the four boxes and the number next to *Trend through the 3 heavy members*.
- For each group, also write down which atom the hydrogens are bonded to in the lightest member.

Tab	Lightest	H bonded to	Actual b.p. (°C)	Trend says (°C)	Gap (°C)	Slope
14						
15						
16						
17						

a. One gap is negative and three are positive. List the three lightest members with positive gaps, and state what their central atoms have in common.

b. All four slopes are positive, including the slopes of the three groups whose lightest member breaks the line. Write one sentence saying what that fact rules out about hydrogen bonding.

c. State the rule this plot supports, in your own words, in a single sentence that accounts for both the slopes and the gaps.

d. Compare the group 16 row against your prediction on page 1. How far apart were your predicted boiling point and the demonstration's *Trend says* value? If you were far off, name the assumption that did it.

The dashed line is an extrapolation, not a measurement. Nobody has ever measured water boiling at $-73\text{ }^{\circ}\text{C}$, because water always has hydrogen bonds. Different reasonable fits put that estimate anywhere from about -70 to $-100\text{ }^{\circ}\text{C}$. What survives all of them is the size of the anomaly: hundreds of degrees, not tens.

4 Working backwards, and where the shortcut breaks

A question rarely hands you a fitted line. It hands you two data points and expects you to build one.

1. Select tab **16** and record the demonstration's *Trend says* value for H_2O together with its *Trend through the 3 heavy members* slope.
2. On paper, build your own two-point line through H_2S (34.08 g/mol , $-60.3\text{ }^{\circ}\text{C}$) and H_2Te (129.62 g/mol , $-2.2\text{ }^{\circ}\text{C}$): find the slope, then use it to predict the boiling point at 18.02 g/mol .
3. Compare the two predictions.

Method	Slope (°C per g/mol)	Predicted b.p. of H ₂ O (°C)	Gap from 100.0 °C
Demonstration, 3-point least-squares fit			
Your 2-point line			

a. The two predictions differ by a few degrees. Does that difference weaken the claim that water is anomalous? Answer by comparing it to the size of the gap.

b. Now the shortcut. Use the data table at the bottom of the page, which lists all sixteen hydrides. Find **two** pairs in which the molecule with the *larger* molar mass boils *lower*. The two molecules in a pair must come from different groups.

c. "Heavier means higher boiling point" worked inside every one of the four series and failed for the pairs you just found. State precisely the condition under which the mass comparison is valid.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and where you were wrong, write down what went wrong.

1. Rank CH_4 , PH_3 and NH_3 from lowest to highest normal boiling point. Their molar masses are 16.04, 34.00 and 17.03 g/mol. Justify the order using intermolecular forces, not molar mass alone.

2. HCl (36.46 g/mol) boils at -85.1°C and HI (127.91 g/mol) boils at -35.4°C . Build the line through those two points and use it to predict the boiling point of HF at 20.01 g/mol. HF actually boils at 19.5°C . Report the predicted value and the size of the anomaly, each to one decimal place.

3. Fluorine is more electronegative than oxygen, and HF has a larger molar mass than H_2O , yet HF boils at 19.5°C and water at 100.0°C . Explain the difference. Your answer should refer to how many hydrogen bonds an average molecule of each can form, and to the number of hydrogens and lone pairs available.

4. CH₄ also misses the line fitted to its own group, by about -37 °C. Explain why that miss is *not* evidence of hydrogen bonding in CH₄, using both the sign and the size of the miss compared with water's.

6 On your own

Nothing here is graded. The demonstration carries more than this worksheet uses.

- Untick **Show extrapolation lines** with all four series on. Using only the shapes of the four measured curves, decide which groups are anomalous. Then tick the box again and see whether you were reading the raw data correctly.
- The tooltip and the data table both give boiling points in kelvin as well as °C. Refit group 16 in K per g/mol. Which changes, the slope or the gap? Explain why.
- The data table has a *Period* column. Sketch the group 16 boiling points against period number instead of molar mass and extrapolate back to period 2. Does water's anomaly survive the change of x-axis, and roughly how large is it now?
- Compare *Fit quality* (R^2 , *heavy members*) across all four tabs. Which series is least linear, and does the ranking of R^2 have anything to do with the ranking of the gaps?